
AI Research Writing Skill: Claim-Evidence Engineering for Agentic Research Paper Generation

Example Paper Package¹

Abstract

AI agents can draft fluent research prose, but scientific writing also requires evidence discipline: claims must be traceable, citations must be verified, figures and tables must become concrete assets, and submission readiness must be checked. This example paper describes AI Research Writing Skill, a cross-platform agent skill for turning ML/AI research repositories into auditable paper packages. The system contribution is a workflow contract rather than a benchmarked model: repository inventory, paper story, claim-evidence map, citation verification, figure and table planning, reviewer self-review, and build checks. We use the repository itself as evidence and show how its templates, references, scripts, and platform entrypoints support a repo-to-paper workflow.

1. Introduction

Large language models and coding agents can accelerate research writing, but prose generation alone is a poor unit of scientific progress. A paper draft is only useful if the reader can inspect which results support which claims, which citations support which background statements, and which artifacts are ready for submission. AI Research Writing Skill addresses this problem by treating paper generation as claim-evidence engineering.

The project builds on a growing ecosystem of research-writing and research-agent resources, including compact paper-writing skills (Master-cai), broad multi-platform writing assistants (Norman-bury), general AI research skill libraries (Orchestra-Research), Nature-style publication workflows (Yuan1z0825), and publi-

cation figure resources (ChenLiu-1996). Its narrower focus is the ML/AI repository-to-paper setting: an agent works inside a codebase, inspects notes and results, and produces a paper package with explicit evidence boundaries.

2. System Design

Figure 2 summarizes the method. The root skill contract defines operating modes for full-paper drafting, story building, section writing, figures, citations, reviewer checks, submission, and automation. References are loaded progressively so that a task uses the narrow guidance it needs rather than the entire documentation tree.

The design has three main principles. First, strong claims must be backed by local repository evidence, experiment artifacts, notes, or verified citations. Second, the output should be durable: story files, claim maps, citation verification notes, figure plans, generated assets, and build checks are written into the target paper repository. Third, completion claims require verification or explicit blockers.

3. Repository Evidence

Table 1 summarizes local evidence used for this example paper. The repository includes venue template directories for major ML/AI conference families, top-level references for writing and submission workflows, helper scripts for static checks and table generation, and cross-platform entrypoints for multiple agent environments.

These counts are not performance metrics. They support only structural claims about what the repository contains. They do not establish that the skill improves writing quality, acceptance probability, or agent reliability.

¹Demonstration Artifact. Correspondence to: Example Maintainer <example@example.com>.

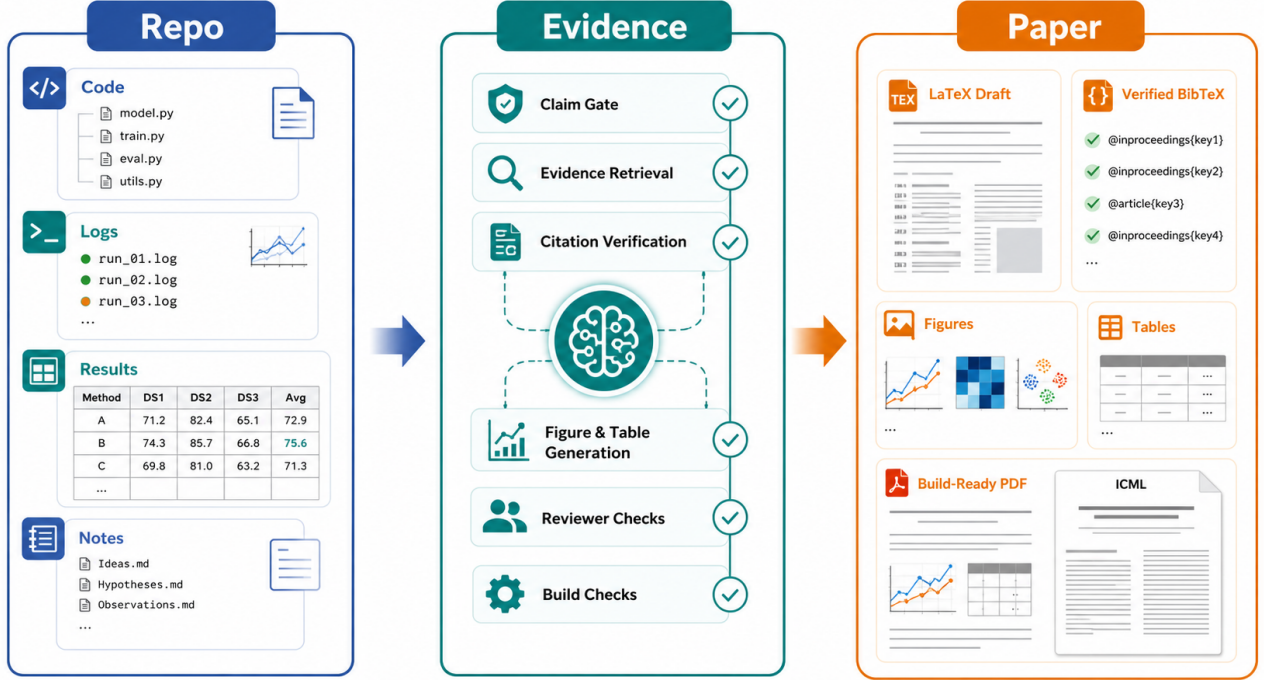


Figure 1. Teaser generated with image generation. The project reframes agentic paper writing as a transformation from repository evidence to a build-ready paper package.

Repository evidence	Count / scope	Paper claim supported
Venue template directories	9	Multi-venue ML/AI template support
Top-level reference files	21	Modular workflow guidance
Top-level script files	8	Deterministic quality checks and helpers
Cross-platform entrypoints	8 groups	Claude, Cursor, Codex, Gemini, OpenCode support

Table 1. Repository evidence used by the example paper. Counts are derived from the local project tree at the time this example was created.

4. Related Projects

Table 2 positions AI Research Writing Skill against neighboring open-source projects. The comparison is intended to clarify scope rather than rank projects.

The key distinction is operational depth in one vertical. Instead of serving as a general thesis assistant or a figure-only resource, this project specifies a complete paper package contract for ML/AI agents working over real research repositories.

5. Limitations

This example paper has important limitations. It does not benchmark the skill against other writing systems. It does not show user-study evidence. It does not prove that generated papers are correct or publish-

able. Venue templates bundled in the repository are convenience copies, so official instructions must still be checked before submission. The skill should therefore be understood as an auditable workflow scaffold, not as a replacement for scientific judgment.

6. Conclusion

AI Research Writing Skill reframes agentic paper drafting as a claim-evidence-engineering process. By combining repository inspection, durable artifact contracts, verified citation practices, figure and table assets, cross-platform packaging, and build-readiness checks, it offers a concrete workflow for turning ML/AI research repositories into auditable paper drafts.

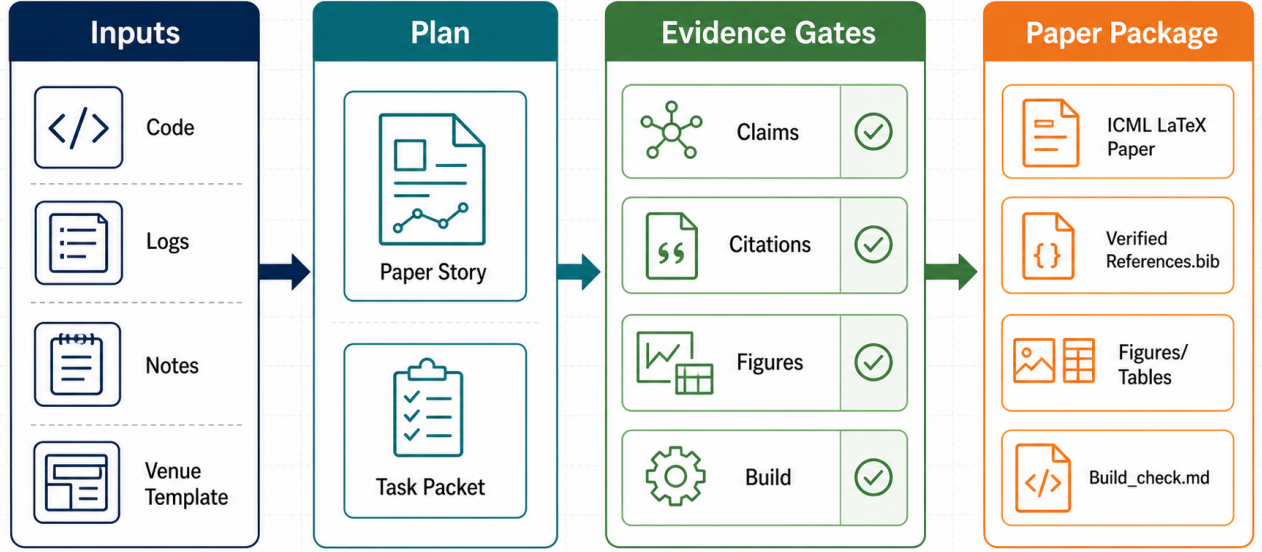


Figure 2. Method overview generated with image generation. AI Research Writing Skill coordinates repository inputs, planning artifacts, evidence gates, and paper-package outputs.

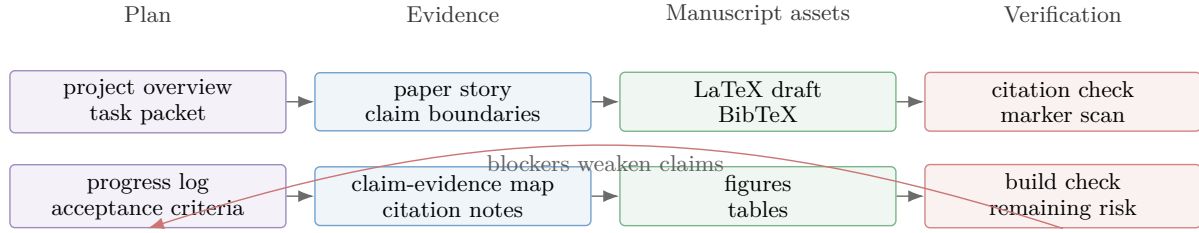


Figure 3. Durable artifact contract used by the skill. Planning artifacts define scope, evidence artifacts constrain claims, manuscript assets carry paper content, and verification artifacts determine whether the agent may claim completion.

References

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Project	Main focus	Distinction of this project
Research-Paper-Writing-Skills	ML/CV/NLP writing skills	Adds repo-to-paper artifacts and checks
research-writing-skill	General research writing	Narrows to AI conference paper packages
AI-research-SKILLS	Broad AI research skills	Focuses on paper execution and readiness
nature-skills	Nature-style publication workflow	Targets ML/AI conference workflows
figures4papers	Publication figure scripts	Connects figures to claims and LaTeX checks

Table 2. Positioning against related open-source projects.